



Lakshya JEE 2.0 (2024)

Determinants

DPP-03

1. Let $f(x) = \begin{vmatrix} \cos x & \sin x & \cos x \\ \cos 2x & \sin 2x & 2\cos 2x \\ \cos 3x & \sin 3x & 3\cos 3x \end{vmatrix}$

Then find the value of $f'\left(\frac{\pi}{2}\right)$

- (1) 0 (2) 2
(3) 4 (4) 6

2. Let $px^4 + qx^3 + rx^2 + sx + t = \begin{vmatrix} x^2 + 3x & x-1 & x+3 \\ x+1 & -2x & x-4 \\ x-3 & x+4 & 3x \end{vmatrix}$

be an identify, where p, q, r, s and t are constants, then the value of s is equal to

- (1) 70 (2) 71
(3) 72 (4) 73

3. Let $\begin{vmatrix} x^2 + 3x & x-1 & x+3 \\ x+1 & -2x & x-4 \\ x-3 & x+4 & 3x \end{vmatrix} = ax^4 + bx^3 + cx^2 + dx +$

k Where a, b, c, d, k are independent of x . Find the value of $4a + 3b + 2c + d$.

- (1) 10 (2) 12
(3) 15 (4) 18

4. Let the following system of equations $kx + y + z = 1$; $x + ky + z = k$; $x + y + kz = k^2$ has no solution. Find $|k|$.

- (1) 0 (2) 1
(3) 2 (4) 3

5. If the system of equations

$$x + 4ay + az = 0$$

$$x + 3by + bz = 0$$

$$x + 2cy + cz = 0$$

have a non-zero solution, then $a, b, c (\neq 0)$ are in

- (1) AP (2) GP
(3) HP (4) AGP

6. The number of all possible values of θ , where $0 < \theta, < \pi$, for which the system of equations $(y + z) \cos 3\theta = (xyz) \sin 3\theta$ $x \sin 3\theta = \frac{2 \cos 3\theta}{y} + \frac{2 \sin 3\theta}{z} (xyz)$

$\sin 3\theta = (y + 2z) \cos 3\theta + y \sin 3\theta$ have a solution (x_0, y_0, z_0) with $y_0 z_0 \neq 0$, is

- (1) 1 (2) 2
(3) 3 (4) 4

7. If $0 \leq \theta \leq \pi$ and the system of equations

$$x = (\sin \theta) y + (\cos \theta) z$$

$$y = z + (\cos \theta) x$$

$$z = (\sin \theta) x + y$$

has a non-trivial solution, then $\frac{8\theta}{\pi}$ is equal to

- (1) 2 (2) 4
(3) 6 (4) 8

8. If $a + b + c = 0$ and $a^2 + b^2 + c^2 - ab - bc - ca \neq 0$, $\forall a, b, c \in R$, then the system of equations $ax + by + cz = 0$, $bx + cy + az = 0$ and $cx + ay + bz = 0$ has

- (1) A unique solution
(2) Infinite solutions
(3) No solution
(4) Exactly two solution

9. If the system of equations $x - 2y + 5z = 3$, $2x - y + z = 1$ and $11x - 7y + pz = q$, has infinitely many solutions, then

- (1) $p + q = 2$ (2) $p + q = 10$
(3) $p - q = 2$ (4) $p - q = 5$

10. The system of equations $ax - y - z = a - 1$, $x - ay - z = a - 1$, $x - y - az = a - 1$ has no solution, if a is

- (1) either -2 or 1 (2) -2
(3) 1 (4) not (-2)

11. The number of values of k for which the system of equations $(k + 1)x + 8y = 4k$, $kx + (k + 3)y = 3k - 1$ has infinitely many solutions, is

- (1) 0 (2) 1
(3) 2 (4) infinite



- 12.** Given $2x - y + 2z = 2$, $x - 2y - z = -4$, $x + y + \lambda z = 4$ then the value of λ such that the given system of equation has no solution is:
 (1) -3 (2) 1
 (3) 0 (4) 3
- 13.** If the trivial solution is the only solution of the system of equations
 $x - ky + z = 0$
 $kx + 3y - kz = 0$
 $3x + y - z = 0$
 Then, the set of all values of ' k ' is:
 (1) $\{2, -3\}$ (2) $\mathbb{R} - \{2, -3\}$
 (3) $\mathbb{R} - \{2\}$ (4) $\mathbb{R} - \{-3\}$
- 14.** Let (x, y, z) be points with integer coordinates satisfying the system of homogeneous equations:
 $3x - y - z = 0$
 $-3x + z = 0$
 $-3x + 2y + z = 0$
 Then the number of such points for which $x^2 + y^2 + z^2 \leq 100$ is
 (1) 1 (2) 3
 (3) 5 (4) 7
- 15.** The system of linear equations
 $x + y + z = 2$
 $2x + y - z = 3$
 $3x + 2y + kz = 4$
 Has a unique solution, is
 (1) $k \neq 0$ (2) $-1 < k < 1$
 (3) $-2 < k < 2$ (4) $k = 0$
- 16.** Let $ax^6 + bx^5 + cx^4 + dx^3 + ex^2 + fx + g =$
 $\begin{vmatrix} (x+1) & (x^2+2) & (x^2+x) \\ (x^2+x) & x+1 & (x^2+2) \\ (x^2+2) & (x^2+x) & (x+1) \end{vmatrix}$. Then,
 (1) $f = 3, g = -5$
 (2) $f = -3, g = -5$
 (3) $f = -3, g = -9$
 (4) None of these



Note: Kindly find the Video Solution of DPPs Questions in the DPPs Section.

Answer Key

1. (3)
2. (2)
3. (3)
4. (3)
5. (3)
6. (3)
7. (3)
8. (2)

9. (3)
10. (2)
11. (2)
12. (4)
13. (2)
14. (4)
15. (1)
16. (4)



PW Web/App - <https://smart.link/7wwosivoicgd4>

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